



COLLEGE
OF
MEDICINE

Miscarriage

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- Sporadic miscarriage is the most common complication of pregnancy.
- The incidence in a clinical recognizable pregnancy is 10–20 per cent.
- The incidence decreases after the 8th week of pregnancy to about 10 per cent and drops to 3 per cent if a viable fetus has been recognized on ultrasound scan



- Maternal age is an independent risk factor for miscarriage.
- Advanced maternal age leads to a decreased number of good quality oocytes, resulting in chromosomally abnormal conceptus leading to a miscarriage.



Recurrent miscarriage

It is defined as spontaneous loss of three or more consecutive pregnancies.

It affects 1 % of all women.
(it will be cover in seminar)



Types of miscarriage

1. Threatened miscarriage
2. Inevitable miscarriage
3. Incomplete miscarriage
4. Complete miscarriage
5. Missed/early fetal demise/ anembryonic pregnancy



Clinical features

1. Threaten miscarriage

Bleeding +/- pain Reassure.

Speculum/pelvic examination – Os
depending on symptoms

Closed

Ultrasound findings:

Intrauterine pregnancy,

i.e. gestation sac with yolk sac +/-

fetal pole and cardiac activity

Action Reassure. Follow up depending on
symptoms



Clinical features

Inevitable miscarriage

Bleeding + pain

Speculum/pelvic examination - Os open

Ultrasound findings:

Intrauterine pregnancy

i.e. gestation sac with yolk sac +-
fetal pole and cardiac activity

Action Loss of pregnancy
inevitable.

Admit and discuss options



Clinical features

incomplete miscarriage

Bleeding +/- pain

Speculum/pelvic examination – Os
open, +/- products at the Os

Ultrasound findings:

Retained products of conception

Action

Admit and discuss options



Clinical features

complete miscarriage

Minimal bleeding +/- pain

Speculum/pelvic examination – Os closed

Ultrasound findings:

Empty uterus or ultrasound appearances showing less than 15 mm in diameter of retained tissue

Action

Reassure and GP follow. β hCG monitoring if ectopic not ruled out



Clinical features

Missed/early fetal
demise/ anembryonic

Pregnancy: bleeding +/- pain Speculum/
pelvic examination – Os closed

Ultrasound findings:

RCOG/RCR criteria CRL >6 mm, no FHM

Empty gestation sac with mean
diameter >20 mm

Action

Expectant/medical/surgical

Management

- There is **currently no evidence** to support the use of the following in the management of miscarriage:
 - ● progesterone supplement for prevention of miscarriage;
 - ● progesterone supplement for treatment of threatened miscarriage;
 - uterine muscle relaxant for prevention of miscarriage;
 - ● bed rest;
 - ● oestrogen supplementation – avoid especially diethylstilbestrol;
 - ● multivitamins

Management – expectant, medical or surgical

- Expectant management
- allows for the avoidance of surgery and general anaesthesia; patients also potentially feel more in control.
- Patients undergoing expectant care are more likely to have retained products of conception or incomplete miscarriage, with a complete evacuation rate of around 30 per cent.
- This is more likely in the presence of an intact gestational sac on ultrasound.
- Comparatively, expectant management is relatively effective in women with incomplete miscarriage, with a complete success rate of around 95 per cent.



Management – expectant, medical or surgical

- Expectant management
- Women undergoing expectant care are more likely to require unplanned surgery due to unacceptable pain and bleeding compared to those who have had surgical evacuation,
- pelvic infection is less common in women managed with expectant care compared to those who had surgery due to the lack of instrumentation of the uterus.



Management – expectant, medical or surgical

- **Surgical management**
- Surgical management or evacuation of products of conception (ERPC) has a high success rate of 95–100 per cent
- vacuum (suction) aspiration was associated with significantly decreased blood loss, less pain and shorter duration of procedure than sharp curettage
- surgical evacuation has its drawbacks, including
- risks such as cervical trauma and subsequent cervical
- incompetence, uterine perforation, intrauterine adhesions or post-operative pelvic infection. The incidence of serious morbidity is about 2 per cent with a mortality of 0.5/100 000.
- The use of prostaglandin vaginally prior to the surgical evacuation can help prime the cervix, make dilatation of the cervix easier and potentially reduce some of the risks and morbidity.



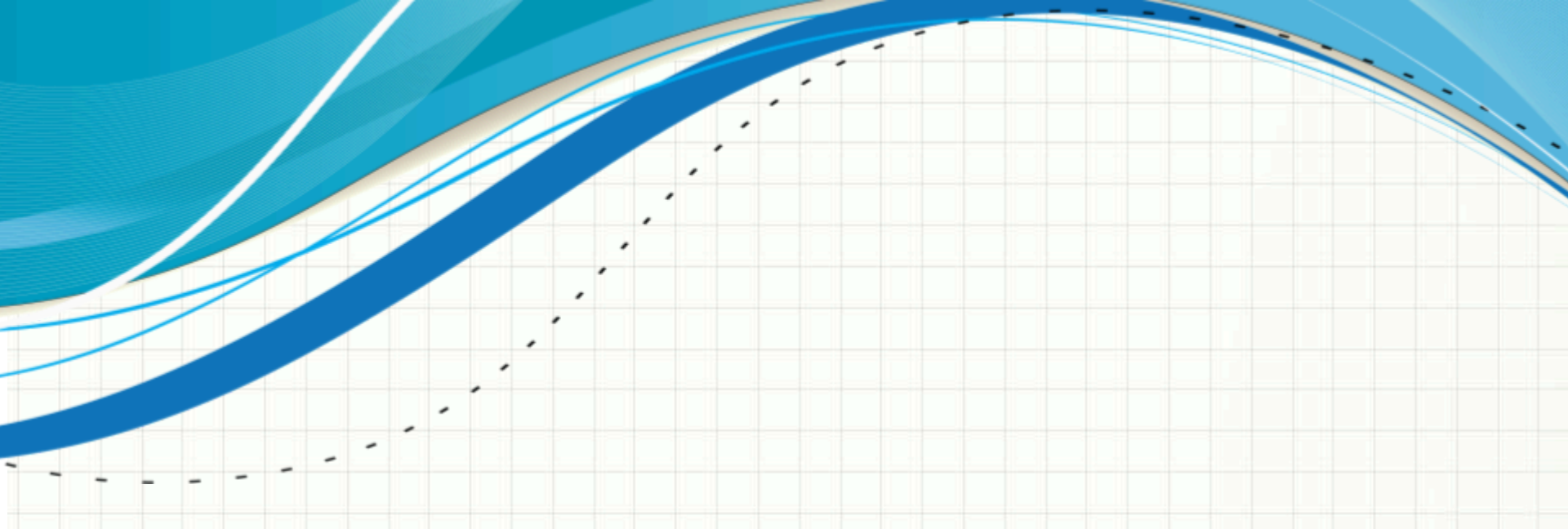
Management – expectant, medical or surgical

- **Medical management**
- Approximately 20 % of women with miscarriage will opt for medical management.
- Prostaglandins are used in single or divided doses administered orally (misoprostol) or vaginally (Gemeprost). Misoprostol is cheap and effective in both oral and vaginal forms.
- medical management is successful in 70–95 % of incomplete miscarriage and in about 50–90 % of delayed miscarriage.
- It has been shown to be most successful in pregnancies under 10 weeks or with a mean gestational sac diameter under 24 mm.



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Appendix